

OPEN SOURCE LAB – ASSIGNMENT

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Class/Batch: CO - 1 / Batch 1 | **Date:** 21/09/2026

Aim

To download all the given datasets from Kaggle, study and compare their licenses and reuse conditions, and then select any one dataset for basic exploration using open-source tools.

Datasets Overview

- Ayurvedic Diabetes Longitudinal Dataset**
 - Source/Platform:** Kaggle
 - File Type:** CSV (.csv)
 - License:** Creative Commons Attribution 4.0 International (CC BY 4.0)
- ACD-MC Dataset V1**
 - Source/Platform:** Kaggle
 - File Type:** CSV (.csv) / ZIP Archive
 - License:** Creative Commons Attribution-ShareAlike 4.0 International (CC BY-SA 4.0)
- Madhumeha Mosaic**
 - Source/Platform:** Kaggle
 - File Type:** CSV (.csv)
 - License:** Creative Commons Attribution 4.0 International (CC BY 4.0)

License Comparison Table

Dataset	License	Attribution Required?	Commercial Use?	Reuse / Modification Conditions
Ayurvedic Diabetes Longitudinal Dataset	CC BY 4.0	Yes	Yes	Free to share, adapt, transform, and build upon for any purpose, provided appropriate credit is given.

Dataset	License	Attribution Required?	Commercial Use?	Reuse / Modification Conditions
ACD-MC Dataset V1	CC BY-SA 4.0	Yes	Yes	Free to adapt and share, but modified versions must be released under the exact same license (ShareAlike).
Madhumeha Mosaic	CC BY 4.0	Yes	Yes	Free to copy, modify, and distribute for personal, academic, or commercial use with proper attribution.

Selected Dataset

Dataset Selected for Exploration: Ayurvedic Diabetes Longitudinal Dataset

Basic Python Code for Exploration

```
import pandas as pd
# 1. Load the dataset
df = pd.read_csv("Ayurvedic_Diabetes_Longitudinal_Dataset.csv")

# 2. Display the first 5 records
print("--- First 5 Records ---")
print(df.head())

# 3. Total number of rows and columns
print("\n--- Shape of Dataset ---")
print("Rows:", df.shape[0])
print("Columns:", df.shape[1])

# 4. Display column names
print("\n--- Column Names ---")
print(df.columns)

# 5. Check missing values
print("\n--- Missing Values Count ---")
```

```
print(df.isnull().sum())
```

Execution Output

The screenshot shows a Jupyter Notebook environment with the following content:

```
[2]: #downloading dataset
!wget -O dataset.zip "https://storage.googleapis.com/kaggle-data-sets/12892739/bundle/archive.zip?X-Goog-Algorithm=GOOG4-RSA-SHA256&X-Goog-Credential=gcp-kaggle-com%40kaggle-161607.iam.gserviceaccount.com/storage.googlegapis.com/kaggle-data-sets/12892739/19799720/bundle/archive.zip?X-Goog-Algorithm=GOOG4-RSA-SHA256&X-Goog-Credential=gcp-kaggle-com%40kaggle-161607.iam.gserviceaccount.com"
!unzip dataset.zip

--2026-09-21 09:27:52-- https://storage.googleapis.com/kaggle-data-sets/12892739/19799720/bundle/archive.zip?X-Goog-Algorithm=GOOG4-RSA-SHA256&X-Goog-Credential=gcp-kaggle-com%40kaggle-161607.iam.gserviceaccount.com
Resolving storage.googleapis.com (storage.googleapis.com)... 142.251.12.207, 172.217.194.207, 74.125.24.207...
Connecting to storage.googleapis.com (storage.googleapis.com)|142.251.12.207|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 3083307 (2.9M) [application/zip]
Saving to: 'dataset.zip'

dataset.zip      100%[=====>]   2.94M  2.27MB/s   in 1.3s

2026-09-21 09:27:53 (2.27 MB/s) - 'dataset.zip' saved [3083307/3083307]

Archive: dataset.zip
Inflating: Dataset_C_7567_CARE_MIXED_COMPACT_FINAL.csv
```

```
[3]: #loading dataset and displaying
import pandas as pd
df = pd.read_csv("Dataset_C_7567_CARE_MIXED_COMPACT_FINAL.csv")

print(df.head())
print("Rows :", df.shape[0])
print("Columns :", df.shape[1])
print(df.columns)
print(df.isnull().sum())
```

	Subject Ref	Centre ID	Assessment_Round	Assessment_Date
0	C01-01533	CTR-01	0	2019/10/28
1	C01-01533	CTR-01	1	2020/02/15
2	C01-01534	CTR-01	0	2021/05/11
3	C01-01534	CTR-01	1	2021/08/10
4	C01-01534	CTR-01	2	2021/11/24

The screenshot shows a Jupyter Notebook with the following content:

```

import pandas as pd
df = pd.read_csv("Dataset_C_7567_CARE_MIXED_COMPACT_FINAL.csv")

print(df.head())
print("Rows :", df.shape[0])
print("Columns :", df.shape[1])
print(df.columns)
print(df.isnull().sum())

```

The output of the code is displayed below the cell:

```

***
  Subject Ref Centre ID Assessment_Round Assessment Date \
0  C01-01533      CTR-01              0      2019/10/28
1  C01-01533      CTR-01              1      2020/02/15
2  C01-01534      CTR-01              0      2021/05/11
3  C01-01534      CTR-01              1      2021/08/10
4  C01-01534      CTR-01              2      2021/11/24

  Months_In_Programme  Days_Since_Last_Review  Age_Yrs  Sx  Ht_m  Wt_kg  ... \
0              0.0              3.6          NaN    59  F    1.66    49.4  ...
1              0.0              2.6          NaN    59  F    1.66    49.0  ...
2              0.0              0.0          NaN    27  M    1.82    79.4  ...
3              3.0              91.0          27  M    1.82    79.1  ...
4              6.5              106.0          27  M    1.82    79.4  ...

  Day_Map  Tobacco_Or_Alcohol  Care_Tier  Escalation_Advised \
0      Yes              No      Tertiary              No
1      Yes              No      Tertiary              No
2      No              Yes      Secondary              No
3      No              Yes      Primary              No
4      No              Yes      Primary              No

  Referral_Required  Programme_Response  QOL_Instrument  QOL_Score \
0      No              Enrolment              NaN              NaN
1      No      Minimal improvement              NaN              NaN
2      No              Enrolment              NaN              NaN
3      No      Minimal improvement              NaN              NaN
4      No      Moderate improvement              NaN              NaN

  Sustained_Response  Programme_Status
0      No              Active
1      No      Discontinued
2      No              Active

```

